

Frontend: Theory Assignment 2

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1. What is CSS Box Model?

The CSS Box Model describes how an HTML element takes up space on a web page. It has four parts: content, padding, border and margin. Content is the actual text or image, padding gives space around the content, border surrounds the padding, and margin gives space outside the border.

2. What is media query? Explain with a sample code.

A media query is used to apply CSS according to the screen size or device. It helps make a website responsive, so the design can change for smaller screens.

```
@media (max-width: 600px) {  
  body { background-color: lightblue; }  
}
```

3. What are the features of Bootstrap?

Bootstrap is a front-end framework which provides a responsive grid system and many ready-made components. It also has utility classes, buttons, forms and JavaScript components. It makes designing responsive web pages easier and faster.

4. Write an example of basic grid structure in Bootstrap.

The Bootstrap grid uses a container, row and columns. For example, three columns can be placed inside one row as follows. The columns divide the available width among themselves.

```
<div class="container">  
  <div class="row">  
    <div class="col">Column 1</div>  
    <div class="col">Column 2</div>  
    <div class="col">Column 3</div>  
  </div>  
</div>
```

5. What is CSS Selectors?

CSS selectors are used to select HTML elements on which we want to apply styles. Some common selectors are element selector, class selector, id selector, universal selector and attribute selector. For example, p selects paragraph elements, .box selects elements having the box class and #title selects the element with id title.

6. What are CSS backgrounds, list the properties?

CSS background properties are used to control the background of an element. The main properties are background-color, background-image, background-repeat, background-position, background-size, background-attachment and the shorthand background property.

7. How can we overlap elements in CSS?

Elements can be overlapped by using the position property. For example, an element can be given position: absolute and moved using top, left, right or bottom. The z-index property is used to decide which element should appear in front.

8. What are the different positioning properties in CSS?

The main CSS positioning values are static, relative, absolute, fixed and sticky. Static is the normal position. Relative moves an element from its normal position. Absolute positions an element according to a positioned parent. Fixed keeps an element in the same place on the screen, while sticky behaves normally until a scrolling point is reached.

9. What is ECMAScript?

ECMAScript is a standard specification for scripting languages. JavaScript is the most commonly used implementation of ECMAScript. It defines features such as variables, functions, objects, classes and other rules of the language.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone?

var, let and const are used for declaring variables in JavaScript. var is function-scoped and can be redeclared and reassigned. let is block-scoped and can be reassigned but cannot be redeclared in the same block. const is block-scoped and cannot be reassigned after it is initialized. The Temporal Dead Zone (TDZ) is the period between entering a block and reaching the declaration of a let or const variable. Accessing the variable during this period gives a ReferenceError.